1. Introduction

1.1 Purpose

We intend to create a comprehensive online course registration and management system that is eligible for Marmara University regulations to overcome the challenges in course management that come from the influence of the pandemic.

1.2 Document Conventions

Conventions have not been used in this document.

1.3 Intended Audience and Reading Suggestions

This document was prepared to introduce the features of the system we designed to our client in a comprehensive way. The second section provides a brief summary of the product's specifications and structure. The third section details the features that should be in the system. The fourth chapter describes the interface of the program, and the last section describes the external requirements.

1.4 Product Scope

This project aims to overcome the challenges of students and advisors in selecting and managing courses during the pandemic period by following the conditions set out in the Marmara University regulation. This system includes functions such as course selection, management of courses received, sending selected courses to advisor approval, managing enrolled students and creating student transcripts.

1.5 References

https://bys.marmara.edu.tr

https://cse.eng.marmara.edu.tr/dosya/eng/cse/documents/general/PrerequisiteTree_2020.pdf

https://www.marmara.edu.tr/dosya/www/mevzuat/2021/mu_yonerge_basari_degerlendirme_2020_v2 04.02.2021.pdf?_t=1612473513

2. Overall Description

2.1 Product Perspective

This software can work independently and totally self-contained. It does not have any graphical user interface. It is designed as an alternative way to other course management systems.

2.2 Product Functions

In this system, students are able to register courses, access their transcripts. After the student completes the course selection process, the system approves or rejects it.

2.3 User Classes and Characteristics

All students and advisors can use this software but students can't access all parts of it.

Students: They will use this software when enrolling to courses in new semesters.

Advisors: Advisors can control their students' course enrolling period. They approve or disapprove students' semester schedule.

2.4 Operating Environment

Operating environment for the course management system is listed below:

- Database: Json files to keep track of students and courses
- Operating system: Windows and Linux
- Platform: Java

Our product is a software produced in java language that runs on the ide console. It can be run on many platforms such as Eclipse, NetBeans, Intellij and Visual Studio Code.

2.5 Design and Implementation Constraints

The most important qualification required to develop the product is to be able to code in the Java language. In addition, the developer should be able to extract, add and remove data from the json file. OOSD is one of the most important elements in our product, therefore it should be written code in this context.

2.6 User Documentation

A detailed user manual should be delivered with the software. The user manual should contain all kinds of details but should not contain any unnecessary information. This document in such a way that a novice user understands both the content and usage of the program should be written.

The developer document, on the other hand, is more geared towards programmers. It will be prepared for them to reach the information they can add.

2.7 Assumptions and Dependencies

At this stage, no third-party applications will be used at this stage, so there will be no dependencies. Code sharing and communication in the project will be via github. In this case, we are dependent on github. Finally, our project is not dependent on software components of any similar project.

3. External Interface Requirements

3.1 User İnterfaces

Our software does not have any user interface, at this stage all operations will occur on the ide console.

3.2 Hardware Interfaces

Since the program is a console application it works in every Java supported ide. Mouse and keyboard are required for use.

3.3 Software Interfaces

In order for the software to work, it is necessary to have Java installed and an IDE that supports Java.

3.4 Communications Interfaces

There is no communication interface in our software.

4. SYSTEM FEATURES

4.1 Access to source code

User can access to all Source code, can change everything by preference

4.1.1 Description and Priority

Users are able to design every part of the source code, if something is wrong, the user can directly change this by a code editor or ide. Users can customize everything by desire. Code is always ready to change and develop. This feature comes with complete control but brings extreme dangers alongside. Users can delete all the software with a small accident, or can make a logic error and it might give them a hard time to fix that. It has extreme plusses and minuses. A bug can be fixed in the moment or a bug which is really hard to fix may be created. All the controls must be used with extreme caution. Users may be only interested in executing the code and don’t care about changing it, the software will run smoothly so its priority is low, but for security it's very high.

4.1.2 Stimulus/Response Sequences

Since all code is in front of the user, when the user writes extra code or changes anything and runs the program every change will be visible at the moment of execution.

4.1.3 Functional Requirements

Users need a computer which has a Java program installed. Software may be executed with ide or command line. Users need a keyboard, without a mouse everything will work but for ease of use we highly recommend a mouse.

4.2 Access to student and course features

4.1.1 Description and Priority

Just like source code, json files which contain properties of students and courses are available to change. Users can change everything about any student or student, they can be deleted, created, changed. This feature is also extremely open for any bad intentions. Users can see everything and can observe anything but the system could corrupt any moment.

4.1.2 Stimulus/Response Sequences

Every change made on a json file will be visible on other json files.

4.1.3 Functional Requirements

The user needs a computer which can edit a json file, a simple text editor which comes alongside with operating systems like Windows, Mac or Linux will be enough.

5. Requirements

5.1 Performance Requirements

The user needs a simple computer which can handle execution of somewhat large Java codes by terminal of the operating system. We highly recommend having a computer that can handle any idea for ease of use and an understandable graphical interface which is somewhat more reliable by non-technical users.

5.2 Safety Requirements

Our information and source code will probably hold in a file, and all information will be there. Our program gives the user all possible permission on everything. The files could be deleted by the user and there won't be any warnings since it's like any other file on the computer. All the information about courses and students may be lost with simple actions. All the information needs to be backed up somewhere, we suggest multiple HDD like devices since they are not connected to any network and portable and easy to protect.

5.3 Security Requirements

The system should ony be accessible to administrators, no strangers should be able to access the system.

5.4 Software Quality Attributes

The software is very light for computers, maintaining it is easy since Java code will work for a long time but still some little changes might be needed from time to time, use of program will require simple knowledge of ide(opening the project and executing it). Since Java code is the same for every computer software will work on any computer with Java program. Maintaining it will be simple since software only contains Java code.

6. Other Requirements

Our software is an open source project and can be used by anyone who wants it. No legal requirements. Software doesn’t need any database, GUI or anything else. A command line Java program will be enough.